# GAP-10L and GAP-10$\psi$: Symmetry Propagation and Metric Stability

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16 July 2026

**AI provenance — Tier B.** AI-assisted; machine-verified against named sources or verifiers; individual attestation is not claimed. Details: `AI_PROVENANCE.md`.

**Abstract**

This note identifies exact sufficient conditions for preservation of the UBT Lorentz slice and for independence of the classical metric from imaginary time. The Lorentz slice is the fixed set of a natural anti-linear involution $\mathcal{J}X = -\overline{X^\sharp}$. Any well-posed evolution whose equations, sources and data are $\mathcal{J}$-equivariant preserves that fixed set by uniqueness. Independently, a $\psi$-flow of the tetrad tangent to a local Lorentz gauge orbit leaves the anticommutator metric exactly unchanged. A $\psi$-translation-invariant well-posed evolution also preserves strictly $\psi$-independent data. These theorems close symmetry-propagation subgaps; verification that the final canonical UBT action satisfies their hypotheses remains open.

The fixed-set propagation argument used below is the standard consequence of equivariance plus uniqueness in a well-posed dynamical system; compare the general symmetry framework for differential equations in Olver [1]. The UBT-specific tasks are to identify the intrinsic involution and to verify that the eventual canonical equations satisfy the required hypotheses.

## 1 The real Lorentz slice as a fixed set

Write a biquaternion as $X = a_0\mathbf{1} + a_k\mathbf{e}_k$ with $a_\alpha \in \mathbb{C}$ and quaternionic conjugation $X^\sharp = a_0\mathbf{1} - a_k\mathbf{e}_k$. Define

$$\boxed{\mathcal{J}X := -\overline{X^\sharp}.} \tag{1}$$

**Theorem 1.1** (Intrinsic characterization of the Lorentz slice)**.** *$\mathcal{J}$ is an anti-linear involution and*

$$\mathrm{Fix}(\mathcal{J}) = \{ix^0\mathbf{1} + x^k\mathbf{e}_k \mid x^a \in \mathbb{R}\} = W_L. \tag{2}$$

*Proof.* For $X = a_0\mathbf{1} + a_k\mathbf{e}_k$, $\mathcal{J}X = -\bar{a}_0\mathbf{1} + \bar{a}_k\mathbf{e}_k$; applying $\mathcal{J}$ twice returns $X$. The fixed-point equations are $a_0 = -\bar{a}_0$ and $a_k = \bar{a}_k$, hence $a_0$ is purely imaginary and the three vector coefficients are real. □

## 2 Equivariance implies dynamical slice preservation

Let $U$ collect the fields needed for a Cauchy formulation, including $\Theta$ and, when necessary, its canonical momentum or first time derivative.

**Theorem 2.1** (Fixed-set propagation by uniqueness). *Assume a Cauchy problem*

$$\partial_t U = \mathcal{F}(U; S) \tag{3}$$

*is locally well posed and unique in a function class invariant under $\mathcal{J}$. Suppose*

$$\mathcal{F}(\mathcal{J}U; \mathcal{J}S) = \mathcal{J}\mathcal{F}(U; S), \tag{4}$$

*and the source and initial data are fixed by $\mathcal{J}$. Then the unique solution remains fixed by $\mathcal{J}$ throughout its interval of existence.*

*Proof.* If $U$ solves Eq. (3), equivariance implies that $\mathcal{J}U$ is another solution. The two solutions have identical initial data and sources. Uniqueness therefore gives $\mathcal{J}U = U$. □

**Corollary 2.2** (Lorentz-slice preservation). *If the complete UBT equations induce a $\mathcal{J}$-equivariant unique evolution for the tetrad variables $E_\mu = D_\mu\Theta/\sqrt{\mathcal{N}_0}$, then initial data $E_\mu \in W_L$ and $\mathcal{J}$-real sources imply $E_\mu(t) \in W_L$ for all times for which the solution exists.*

## 3 Imaginary-time metric stability

Let ${e_\mu}^a(x, \psi)$ be the real tetrad coefficients and $g_{\mu\nu} = {e_\mu}^a {e_\nu}^b \eta_{ab}$.

**Theorem 3.1** ($\psi$-vertical Lorentz flow leaves the metric invariant). *If*

$$\partial_\psi {e_\mu}^a = {\Lambda_\psi}^a{}_b {e_\mu}^b, \qquad \Lambda_{\psi ab} = -\Lambda_{\psi ba}, \tag{5}$$

*then*

$$\boxed{\partial_\psi g_{\mu\nu} = 0.} \tag{6}$$

*Thus strict $\psi$-independence of the tetrad is sufficient but not necessary: its $\psi$ dependence may be pure local Lorentz gauge.*

*Proof.* Differentiate $g_{\mu\nu}$. The two terms are $\Lambda_{\psi ab}{e_\mu}^b{e_\nu}^a$ and $\Lambda_{\psi ba}{e_\mu}^b{e_\nu}^a$, which cancel by antisymmetry. □

**Theorem 3.2** ($\psi$-translation symmetry propagation). *Suppose the complete Cauchy/boundary problem is invariant under $\psi \mapsto \psi + s$ and has a unique solution. If all initial and boundary data and sources are $\psi$ independent, then the solution is $\psi$ independent.*

*Proof.* For every $s$, the translated field $U_s(x, \psi) = U(x, \psi + s)$ is a solution with the same data. Uniqueness gives $U_s = U$ for all $s$, hence $\partial_\psi U = 0$. □

## 4 Status

Defensible status

**GAP-10L-SYM: CLOSED CONDITIONALLY [L1].** The Lorentz slice is the fixed set of the intrinsic involution $\mathcal{J} = -\overline{(\cdot)}^\sharp$, and every unique $\mathcal{J}$-equivariant evolution with fixed data preserves it.

**GAP-10L-DYN: NARROWED.** It remains to prove that the finalized UBT action, its paired connections, sources and chosen well-posed formulation satisfy the equivariance and uniqueness hypotheses.

**GAP-10$\psi$-KIN: CLOSED [L1].** A $\psi$-flow tangent to the local Lorentz gauge orbit leaves the metric exactly invariant.

**GAP-10$\psi$-SYM: CLOSED CONDITIONALLY [L1].** A unique $\psi$-translation-invariant evolution preserves $\psi$-independent data.

**GAP-10$\psi$: NARROWED.** The canonical action must still show that the physical classical branch obeys one of these sufficient mechanisms and that no unstable non-gauge $\psi$ excitations contaminate the observed metric.

## References


[1] P. J. Olver, *Applications of Lie Groups to Differential Equations*, 2nd ed., Springer, New York (1993).